

Admitting One Free Field Cuts a CMDB's Measured Value Nearly in Half

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Abstract

Configuration management database (CMDB) programmes are justified by the analytics they enable, and that justification is a comparison against a baseline the analyst chooses. On a public event log of 45,455 incidents from a bank's IT operation whose affected-item field is 100% populated, we show the choice is worth nearly as much as the data itself. Predicting at intake whether an incident will be reassigned, knowing the affected configuration item is worth +0.183 AUC against four intake fields, and +0.103 [+0.094, +0.113] once one more free field is added: which group logged the incident. The gap survives three estimator families, including one in which item identity is a single column rather than 2,554, and holds at three target thresholds; on two further targets it is directionally consistent but unresolved. Stated as a detection problem it is larger than the AUC ratio of 1.8 suggests — at a review capacity of 5% of arrivals, omitting the field credits the CMDB with 4.3 [3.2, 6.4] times as many additional catches. Two measurements explain the overlap: the opening group adds under 0.01 AUC to a model that already knows the item, and the shrinkage is graded in the field's resolution, running 27% to 44% from one bit to 49 levels — a consistency check rather than an independent prediction. We also report three things we got wrong: a null that could not fail, an operational factor we first reported without an interval, and a field whose meaning we mis-stated in an earlier draft.

1 Introduction

A CMDB programme is expensive and is justified by what it will enable. That justification rests on a comparison whose second arm — performance without configuration data — is a choice the analyst makes, usually silently, by deciding which already-available fields to include.

On one organisation's incident log the choice halves the answer, and we can say exactly where the difference goes. This is a short paper with one result and one mechanism.

What kind of contribution this is. We report no deployed system. This is a retrospective study on a public benchmark, offered as an evaluation practice: the quantity we measure is not a model's accuracy but the number a deployment decision turns on, and that number is set by a field-admission choice rarely written down. Section 6 states what it costs in the units a desk operates in.

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On what this paper used to claim. Earlier versions reported larger findings withdrawn under adversarial review, each an artifact of a control we built ourselves. Three corrections survive here as reported results rather than silent edits, because each is the kind of error this paper is about: a null that could not fail (Section 5), a factor reported without an interval (Section 6), and a central field whose meaning we asserted rather than checked (Section 3).

2 Background

The CMDB and what justifies it. A CMDB records configuration items and their relationships, and is the asset on which the configuration-management practice of ITIL and its successors rests. Survey work on ITSM framework adoption (Marrone and Kolbe 2011) finds that realised benefits grow with implementation maturity, but measures them as practitioner-reported outcomes rather than as the analytic value of the configuration data itself. We are not aware of a peer-reviewed measurement of what knowing the affected configuration item is worth to a named prediction task; the material addressing that question directly is vendor documentation. The gap is the occasion for this paper, and part of the reason it exists is that the data to close it barely does: of the three public ITSM incident logs in common use, one records the affected item on every incident, the second records it on 0.2%, and the third has no such field at all (van Dongen 2014; Amaral, Fantinato, and Peres 2018; Steeman 2013). That is also why this study is single-organisation, and it is a constraint rather than a choice.

The methods a CMDB is bought to enable. Surveys of AIOps for failure management (Zhang et al. 2024; Remil et al. 2024; Notaro, Cardoso, and Gerndt 2021) catalogue triage, failure-prediction and root-cause methods that take configuration data as input. Incident triage in particular — deciding which team owns an incident — has been treated as a learning problem on operational logs at industrial scale (Chen et al. 2019), as has failure prediction across a cloud estate (Lin et al. 2018).

Prediction on incident event logs. Our task sits in predictive process monitoring, surveyed and benchmarked across real-world logs (Teinemaa et al. 2019; Verenich et al. 2019) on data of the kind process mining formalises (van der Aalst

2016). The BPI Challenge logs are the standard public instances (van Dongen 2014; Steeman 2013); comparable exports exist elsewhere (Amaral, Fantinato, and Peres 2018). Such logs carry documented quality hazards (Suriadi et al. 2017), including values denormalised from a record onto its events, which we rule out below. Two lines bear directly on our field: the opening group is a resource attribute rather than a case attribute, and the intra- versus inter-case distinction that makes such features awkward is treated by Senderovich et al. (2017); and establishing what a resource stamp on an event actually denotes is organisational mining (van der Aalst, Reijers, and Song 2005), which is what Section 3 does from first principles.

What the comparison is measured against. The value of a feature *given* a baseline is what algorithm-agnostic variable-importance frameworks formalise (Williamson et al. 2021), and our overlap is the interaction additive importance measures handle (Covert, Lundberg, and Lee 2020). That measured importance depends on what the model already has is therefore not news; what is not established is its magnitude for configuration data on a named operational task, or that the field it overlaps with is one the organisation already records for nothing. Rendle, Zhang, and Koren (2019) and Ferrari Dacrema, Cremonesi, and Jannach (2019) show inadequately specified baselines accounting for reported gains in recommender systems; benchmark conclusions are sensitive to variation that is seldom reported (Bouthillier et al. 2021), and documentation of the choices producing a result is thin (Gundersen and Kjensmo 2018). Where the choice concerns which fields may enter, it shades into leakage (Kaufman et al. 2012; Kapoor and Narayanan 2023), which for predictive process monitoring needs explicit construction to avoid (Weytjens and De Weerd 2022). We report this failure mode for a deployment decision rather than a leaderboard, where field admission is not a modelling detail but the number the business case turns on.

3 Data and Task

We use the incident records of the BPI Challenge 2014 log (van Dongen 2014) from Rabobank Group ICT, recorded in HP Service Manager, with the activity log shipped alongside.

Cleaning. Of 46,809 rows, 203 carry no incident identifier and one an unparseable timestamp. A further 1,150 opened before 2013-10-01 are left-censored — already open when the extract began — and are reassigned at 81.2% against 40.0% for the incidents we keep. Removing them leaves 45,455 incidents to 2014-03-31; a temporal 70/30 split gives 31,818 training and 13,637 test, positive rates 0.413 and 0.372. The cutoff is not taken from the outcome: monthly volume rises from 857 incidents in September 2013 to 8,606 in October. The extract is right-censored too — incidents opened near its end cannot have long lives — so we re-measured on cohorts truncated up to a month early; the shrinkage moves between 42% and 45%, so the boundary does not drive it. The affected-item field is 100% populated across 2,929 items, 2,554 seen in training.

Task and estimator. Predict at creation whether an incident will later be reassigned. We call the target reassignment rather than misrouting throughout: it fires on 37.2% of test incidents, which is routine handling and not error, and Section 6 reports the ladder at stricter thresholds. Median handling time rises from 1.59 hours for incidents never reassigned to 47.53 hours for those reassigned five or more times; we use this to motivate the task, not as a recoverable saving. The primary estimator is one-hot logistic regression, chosen because it is bin-free; two further estimator families are reported in Section 4. Encoders and rankings are fit on training data only; intervals are 2,000-resample paired bootstraps, except Table 2, whose counts are re-ranked in each draw and use 400. Monte-Carlo nulls use 15 to 100 draws, the count set by cost and recorded in the code that produces each.

The free field, and what it is not. The activity log records, on the `Open` event of every incident, an assignment group: 49 in training, none missing. An earlier draft of this paper called it the queue the service desk routed the incident to. That was an assertion, not a finding, and it is wrong. Three checks say so. The `Open` rows carry 50 distinct groups where the `Assignment` rows carry 218, and a routing destination cannot be less diverse than the teams that then do the work. One group accounts for 67.0% of `Open` rows but only 18.4% of all activity rows. And the `Open` group matches the group on the incident’s first `Assignment` activity for just 15.1% of the 37,577 incidents that have one. The field is the group that *logged* the incident: one central desk for two thirds of tickets, a long tail for the rest. We do not characterise that tail — its openers match the first `Assignment` group only 21.8% of the time and appear anywhere in the later work rows only 59.1%, so calling them teams opening their own work would be another assertion of the kind we are correcting. It is also concentrated and unstable: the largest group holds 62.1% of training incidents, and the number of live groups falls from 49 to 32 between our two halves.

We report the error because the correction carries the paper’s method. Nothing measured changes; what changed is a claim about what the field *meant*, made on no evidence, in the way this paper accuses business cases of treating baselines.

Why not the real routing decision. The obvious repair — use the first `Assignment` group — is not available: it occurs strictly after `Open` on every incident that has one, a median of 46 minutes later, and 7,878 of 45,455 never have one. It fails the criterion below, and substituting it would introduce the leakage that criterion prevents.

Admissibility, and mutation. A field on the `Open` row is admitted only if it is a per-event observation, tested by whether it varies across an incident’s activity rows; the opening group varies for 92.56% of cohort incidents and matches the last-observed group for only 21.35%. Because the incident file is a closed-record snapshot we also report how often each field is edited after creation: Urgency on 1,107 incidents (2.44%), Impact on 1,084 (2.38%), the affected item on 159 (0.35%). Incidents whose Impact was edited are reassigned at 0.66 and those whose Urgency was edited at 0.67, against

Baseline	AUC	+ item id.	gain [95% CI]
intake only	0.562	0.746	+0.183 [+0.172, +0.195]
+ opening group	0.644	0.748	+0.103 [+0.094, +0.113]

Table 1: The value of knowing the affected configuration item, with and without a field the service desk already records.

0.39 for the rest, so this is not neutral; Section 4 reports the headline with those fields dropped and with the edited incidents removed.

4 The Result

Against four intake fields the affected item looks decisive. Adding the opening group cuts its measured value by 44%. The gains are roughly 28 and 17 standard deviations outside a null in which the incident-to-item association is shuffled, preserving column count and marginal distribution while destroying the signal — we round because re-drawing the same null at a different draw count moves the first figure by about 0.7 — a comparison that matters because adding item identity adds 2,554 indicator columns, which is not free under a fixed penalty. These z figures pool the null’s Monte-Carlo dispersion with the gain’s own bootstrap standard error; dividing by the null alone, as an earlier draft did, would report 51 and 33.

It is not a split artifact: across six split points the two gains range +0.175 to +0.194 and +0.091 to +0.119 (Figure 1b), and across split point, target threshold and cleaning cutoff the second ranges +0.067 to +0.130 — wider than any bootstrap interval, which conditions on all of those. Nor does the disclosed field editing explain it: reducing the intake block to Category alone gives +0.106, and restricting to the 44,227 never-edited incidents gives +0.107.

It is not an artifact of the estimator. The dimensional null rules out the regularisation burden of 2,554 sparse columns, but a reader may want the effect shown where that burden cannot arise. We re-represent item identity as a single cross-fitted target-encoded column — fitted on training only, out-of-fold, so no row sees its own outcome — which makes the item one continuous feature and makes histogram boosting usable on a variable that otherwise collapses 2,554 items into 137 bins. Across one-hot logistic regression, logistic regression on the encoded item, and boosting on it, the first rung ranges +0.173 to +0.184, the second +0.091 to +0.104, and the shrinkage 42% to 48%. Encoding a *shuffled* item column under the same pipeline returns -0.0001 ± 0.0016 and $+0.0042 \pm 0.0020$. The first collapses to zero; the second does not quite, and we report it rather than round it away because it bounds how much of the boosting rung could be encoding artifact.

5 Where the Difference Goes

The two rows differ by 0.080, which is not itself informative: for any two variables, the drop in one’s gain when the other joins the baseline equals the drop in the other’s. Read four

ways, the item is worth +0.183 over intake and +0.103 once the group is present; the group is worth +0.082 over intake and +0.002 once the item is present. Two measurements explain the overlap.

First: adding the opening group to a model that already knows the item is worth +0.0017 [+0.0001, +0.0034]. It clears a matched-dimension null of -0.0009 ± 0.0005 that no draw reaches, so it is not the cost of adding columns; but across split points and penalties it ranges +0.0002 to +0.0072, so the honest statement is that its unique contribution is *under 0.01 AUC and not resolvable more finely*.

Second: the group’s predictive content is largely a function of the item. Randomising the group label *within* each item — which destroys the label’s own identity while preserving what the item implies about it — retains 91% of the group’s +0.082 gain.

A floor that could not fail. An earlier draft compared that 91% against a floor of 2% and claimed a margin of 89 points. The floor was wrong. It drew cell labels independently *per row*, so two incidents sharing an item landed in different cells and the association was destroyed by construction: the null could only ever return zero. Rebuilt as a random partition *of items*, which preserves item-level grouping and knows nothing about who opened anything, the floor rises with the partition’s fineness — 41% at 49 cells, matching the field’s own cardinality, 77% at 400 (Figure 2b). It is a 30-draw Monte Carlo and its dispersion is wide — 12 points at 49 cells — so the margin below is indicative, not resolved. It is matched on cardinality but not on mass, and mass-matching moves a floor by tens of points in our reverse-leg analysis, so it bounds the margin from one side only. Note what this sweep cannot show: at one cell per item the null *is* the real leg, so “the real leg beats every floor” would be true by construction. The informative comparison is at matched cardinality, where a grouping that knows nothing about intake retains 41% against the real 91% — a margin of 50 points, not the 89 we published against a null that could only return zero.

How far this goes, and where it stops. Without any estimator, item identity carries 60.4% of the opening group’s information and the group carries 19.6% of the item’s. Both are plug-in estimates over thousands of conditioning levels and inflated by finite samples — shuffling item identity leaves floors of 14.0% and 4.5% — so read the asymmetry, which survives with 46 points against 15, not the levels. Nor is it determination: a modal-group lookup on item identity reproduces which group logged the incident on 90.3% of test incidents against 78.6% for always guessing the largest, and class-balanced reaches only 34.1%. Item identity resolves the coarse contrast — central desk or specialist team — and much less of the fine one.

The shrinkage is graded in the field’s resolution. That reading implies a consistency check, and it passes. Coarsening the group and re-running the ladder gives shrinkages of 27%, 28%, 36% and 44% at two, four, eleven and 49 levels. A check rather than a prediction, because at each resolution the shrinkage is close to that coarsened field’s own gain

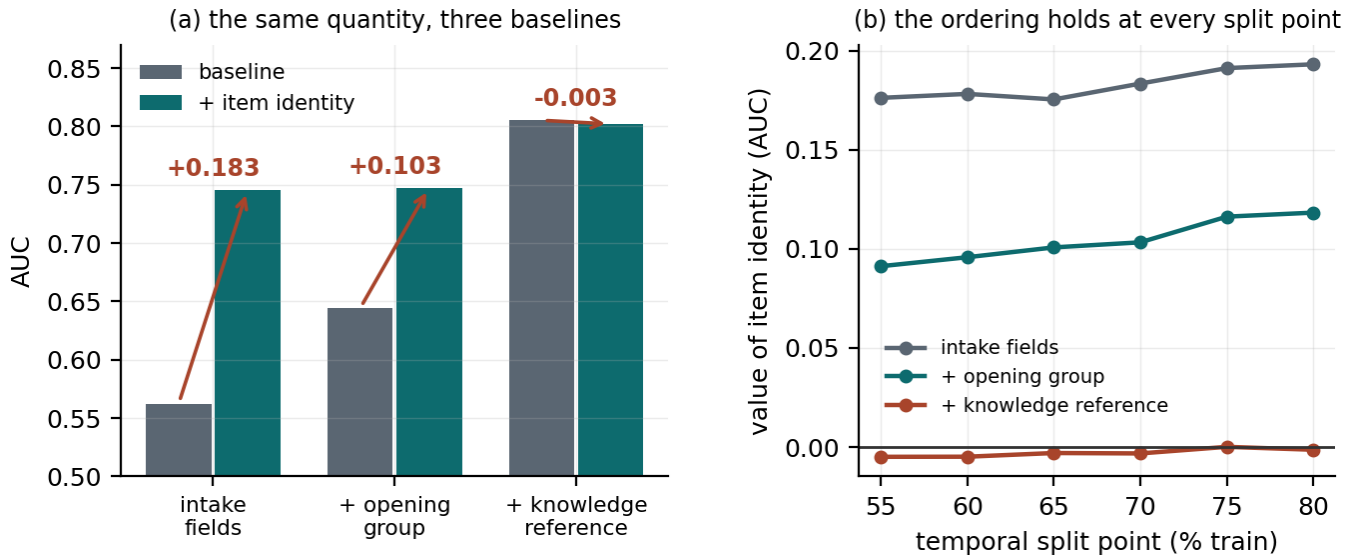


Figure 1: (a) The same quantity under three baselines; the third is discussed in Section 8. (b) The ordering holds at every temporal split point from 55% to 80%.

over intake — by the identity above, the same statement as the first measurement — and the two coarsest points have overlapping intervals, so only the trend is claimed. A single binary flag — central desk or not — already recovers 61% of the group’s baseline gain and 63% of the shrinkage. The field that must be admitted is close to one bit, far cheaper to reproduce elsewhere than a 49-way taxonomy.

Is the field really free? One objection goes to the heart of the reading. The opening group may be free only because a human at the desk already knew what the ticket was about — in which case it is undocumented configuration knowledge, and a baseline containing it is not a CMDB-free baseline. The one-bit result makes this concrete: the contrast doing most of the work is the central desk against everything else, and Section 3 declines to say what everything else is, so we cannot say whether the knowledge sits there. The measurement is the same under both readings; the moral is not. If the group is independent, a CMDB is worth half what the naive comparison says. If it is tacit configuration knowledge, our +0.103 is a lower bound and the organisation already holds much of what the CMDB would tell it, unpaid for. The log cannot identify which is true, so we choose neither. What survives either way is that the business case cannot be written against the intake block alone.

What we do not claim. The reverse framing — that the item column “stands in for” the group — is not established. We measured it, by randomising items within groups, and obtained 44% of the item’s gain — but a routing-blind equal-size partition of items retains *more*, 56%, and one whose cell sizes follow the group’s own mass profile retains 25%. The leg is floor-dominated and we exclude it. (An earlier draft reported a negative value for the second floor. That construction drew cells per row, the same defect described above, and the equal-size floor was computed from different

partitions for training and test; both are corrected here.) Nor do we claim a direction of causation: the log records no counterfactual, so whether the item drives who logs the ticket or both follow a third cause is not identified here. Our claim is about what an analyst’s baseline choice does to a measured number, which does not depend on which is true.

Two fields we did not admit, and why. Our baseline is not a terminus either. Hour of day and day of week are free and creation-time; adding both moves the item’s value from +0.103 to +0.099. Service Component WBS (aff) is harder: 100% populated, free, and admitting it takes the measured value to +0.023. We exclude it as near-deterministic in the item — only 58 of 2,929 items carry more than one value — so it is configuration data under another name. That criterion cuts both ways: 2,060 of 2,554 training items also map to a single opening group, though those items carry just 8.8% of incidents, so the high-volume ones are the multi-group ones. We know of no principled threshold that admits the group and excludes the service component. A reader who draws the line differently should read our headline as +0.023 — which is the decision this paper is about, and why we report a ladder rather than a number.

6 What the Difference Buys

An AUC difference is not a quantity a service desk can act on, so we state the same comparison as detection. Fix a review capacity — the share of arrivals a desk can look at before routing — rank the test incidents by predicted risk, and count how many reassignment-bound incidents each model surfaces. Both arms use matched models: the honest contrast adds item identity to intake-plus-group, the naive one to intake alone, mirroring Table 1. An earlier draft did not match them, crediting item identity with the group’s own

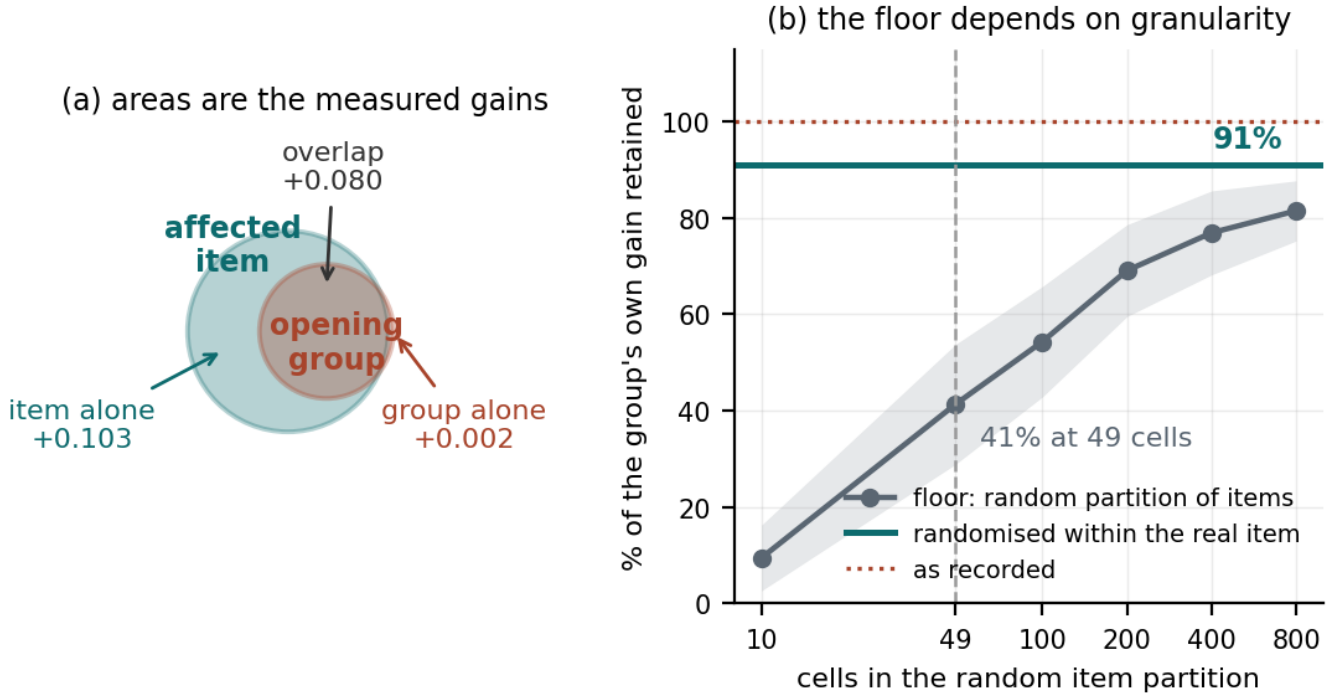


Figure 2: (a) The overlap, drawn to scale: each circle’s area is that field’s measured gain over the intake block, and the centre distance is solved so that the intersection equals the measured overlap. The opening group lies almost entirely inside the item. (b) Randomising the group label within each item costs little; the same randomisation within a random partition of items costs more, by an amount that depends on how fine the partition is.

Capacity	honest extra	naive extra	factor
5%	+63 [43, 84]	+271 [242, 300]	4.3 [3.2, 6.4]
10%	+37 [6, 70]	+361 [304, 404]	9.8 [5.5, 39.9]
20%	+28 [-28, 82]	+470 [407, 524]	16.8 [5.8, 221.0]

Table 2: Extra reassignment-bound incidents surfaced by adding item identity, against a baseline that includes the opening group (honest) and one that omits it (naive). Paired bootstrap over test rows, with ties at the capacity cut broken at random in every draw. Point estimates and intervals are both taken over the same 400 draws.

contribution.

At 5% capacity, knowing the affected item surfaces 63 [43, 84] more reassignment-bound incidents than the intake-plus-group baseline does, or about 34 a month at this log’s volume. An analyst who had omitted the group would have credited item identity with 271 [242, 300]. The omission overstates the operational gain by a factor of 4.3 [3.2, 6.4], against a ratio of only 1.8 between the same two gains measured as AUC.

The factor grows with capacity — 9.8 at 10%, 16.8 at 20% — but so does its interval: the honest denominator shrinks to a few dozen incidents, at 20% it includes zero, and both of those factor intervals are conditioned on a positive denominator, which the 5% row does not need. We quote the 5% figure and claim only that the overstatement is several-

fold and larger than the AUC ratio implies. Two things make it larger. AUC averages over every operating point while a capacity-limited desk works at one, where the group-aware baseline is near its ceiling. And the naive baseline cannot rank finely: four low-cardinality intake fields emit only 23 distinct scores over 13,637 test incidents, so at 5% capacity just 47 rows sit strictly above the cut and the rest come from a tied block of 1,944. That is a property of the baseline an analyst would use, not an artifact of ours, but it makes the factor partly about resolution rather than information. None of this says a surfaced incident is a corrected one.

Three target definitions. Because reassignment is common, we repeat the ladder at stricter thresholds. Requiring two or more reassignments (21.8% of incidents) gives +0.131 and +0.068; three or more (10.6%) gives +0.151 and +0.080. The magnitude is threshold-dependent and not monotone, but the interval on each +group rung excludes zero (we compute intervals for that rung only), the ordering never reverses, and the shrinkage stays between 43% and 49%. The transferable quantity is the shrinkage, not the gain.

7 Does It Survive a Change of Task?

A gap on one target may be a property of it, so we repeated the ladder on two further outcomes. *Reopening* is close to an independent failure mode: 2,096 incidents, correlating with reassignment at +0.14. *Long handling* — handle time above the training third quartile — correlates at +0.40, so it is

corroboration only.

On reopening, item identity is worth +0.083 against the intake block and +0.055 once the group is admitted, a reduction of 33%; on long handling, +0.118 and +0.078, a reduction of 34%. Every rung sits outside its matched-dimension null. But the shrinkage is what we call transferable, so it needs its own interval, and paired bootstraps give [40, 48] on reassignment, [28, 38] on long handling and [-1, 60] on reopening. On the one target we argue is near-independent the shrinkage is *not* resolvably different from zero ($P \leq 0$ is 0.03), and long handling, correlated at +0.40 with the primary target, cannot carry a replication alone. The effect is directionally consistent on both and resolved on neither of the two. Reopening is rare and its gains stand only 5.5 and 4.2 pooled standard deviations from their nulls. The reduction is largest on the target the opening group is most directly about — consistent with Section 5 rather than predicted by it: the mechanism says nothing about how shrinkage should order across targets. That is why we state the transferable claim as the gap’s existence rather than its magnitude.

8 One Thing We Could Not Establish

A third field, the knowledge-article reference, sits on the same Open row, is 100% populated, and costs nothing. Adding it to the baseline raises AUC to 0.805 and drives the measured value of item identity to -0.003. Whether that figure belongs in Table 1 depends on whether the field is available at creation, and we could not determine this.

Two structural tests both fail. A field that varies within an incident is certainly a per-event observation; the knowledge reference never varies — but neither does *Interaction ID*, which has 45,426 distinct values for 45,455 incidents and is plainly a creation-time key, so constancy is evidence of granularity, not timing. Exact identity with the closed record fares no better: the knowledge reference matches its closed-record column for 100.000000% of incidents, but so does *Interaction ID* for 99.997628% of the 42,151 single-interaction incidents — one mismatch against a pass mark of 100.000000%.

We therefore make no claim about this field. The -0.003 is also inside the dimensionality null of Section 4 and changes sign under penalty tuning, so it is not resolvable on either ground. This is the paper’s thesis turned on itself, and that reading is correct: our headline is contingent on a field-admission decision we cannot adjudicate. The transferable part is the negative — a field being free, populated and present on an opening record does not make it creation-time. The collection ships an interaction detail file we did not obtain; if it records this field at creation the question is answerable, and our inability to settle it is a limit of the three files we hold, not of the export.

9 Estate Concentration

Required CMDB scope is bounded by a property of the estate needing no model: the top 8 items generate 30.2% of training incidents, the top 64 70.5%, and the top 128 — 5.0% of the training vocabulary — 82.0%, all from a frequency count before funding anything.

Identifying only those items recovers most of it. Averaged over five split points, the top 8 recover 56% of the +0.103, the top 64 recover 88% and the top 128 recover 93%; across those splits the three range over 52–58%, 81–93% and 90–96%. These are min–max spreads over a design choice, not bootstrap intervals, and we write them as ranges to keep the distinction. We give the mean and range rather than one split’s curve because a single curve is not monotone at this resolution: at $k = 32$ the across-split spread is 9 points. The same holds with the group removed from the baseline — 89% at $k = 64$ — so this is a property of the estate’s concentration, not of our field-admission decision.

10 Limitations

One organisation, of necessity rather than choice (Section 2); one platform; six months of 2013 and 2014 data from an on-premises product whose estate and practice have moved since. The values +0.183 and +0.103 are properties of this estate; what transfers is that the gap exists, that it is large, and that the mechanism runs from item to group. The reverse — the item standing in for the group — is the leg Section 5 excludes. Reassignment is a proxy for misrouting and also fires on legitimate escalation; the within-item shuffle preserves item membership but not other correlates of identity, so it bounds the proxy share rather than isolating it; and the opening group’s distribution drifts between our halves.

11 Conclusion

Knowing which configuration item an incident concerns is worth +0.183 or +0.103 AUC for predicting reassignment, depending on whether the baseline includes one free field: which group logged the ticket. Stated as detection at a fixed review capacity, the same choice moves the credited benefit several fold. The difference is an overlap — a model given the item has already been told most of what the group would say — so a business case built on the first figure is measuring a CMDB plus a field the organisation was already recording for nothing.

Acknowledgements and Reproducibility

This work benefited from the review and domain challenge of a practitioner in IT service management. The dataset is public (van Dongen 2014); code, figures and a checker that recomputes every quantity here: <https://github.com/sudhirdixit1/cmdb-routing-queue-baseline>.

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